

Abhinandan S Prasad

Curriculum Vitae

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My research theme is to employ mathematical and machine learning methods to address the system problems in emerging Cloud Networking technologies such as Systems ML, Algorithmic Game Theory, Edge-Cloud Continuum, and 6G.

Summary

- Computer Science researcher with 10+ years of industry and academic experience in emerging Cloud Networking technologies.
- Analytical and innovative thinker with leadership and problem-solving skills.
- 10+ publications in Tier 1 conferences and high-impact journals (**360** Citations). My Google Scholar profile can be found [here](#).

Education

- Jun 2015 – **Ph.D.**, *University of Göttingen, Germany*
Jun 2018
 - Thesis Topic: *Automated Resource Provisioning of Fairly Priced Resources*.
 - *Magna cum laude* (graduation with great honors)
 - Early stage researcher (ESR) fellow in Marie Curie ITN project "CleanSky"
- July 2009 – **Master of Technology (M.Tech)**, *International Institute of Information Technology Bangalore, India*
Jun 2011
 - Thesis Topic: *A Mechanism Design Approach to Resource Procurement in Cloud Computing*.
- Sept 2002 – **Bachelor of Engineering (B.E)**, *The National Institute of Engineering, India*
Jun 2006
 - Project: *Machine Dependent Optimization for Intel Strong ARM Processors*.

Professional Experience

- Jun 2025 – **Assistant Professor**, *IIT Palakkad, India*
Current
- Aug 2023 – **Assistant Professor**, *IIT Ropar, India*
Jun 2025
- Mar 2020 – **Associate Professor**, *NIE Mysuru, India*
Aug 2023
- Jul 2018 – **Assistant Professor**, *NIE Mysuru, India*
Mar 2020
- Jun 2015 – **Early Stage Researcher**, *CleanSky ITN, Germany*
Jun 2018
- Aug 2014 – **HCL Engineering and Research Services**, *Bangalore, India*
Apr 2015
 - **Technical lead**: Lead wireless management systems (WMS) platform team (WTK).
- Oct 2013 – **Alcatel Lucent India**, *Bangalore, India*
Jul 2014
 - **Lead Engineer**: Lead wireless management systems (WMS) platform team (WTK).

- Aug 2011 – **Motorola Mobility, Bangalore, India**
 Oct 2013 ■ **Software Engineer:** Worked on high-density, fully redundant Motorola BSR64000 CMTS/Intelligent edge router.
- Sep 2008 – **Software Paradigms India, Mysore, India**
 Apr 2009 ■ **Trainee Software Engineer:** Migration of legacy point of sale PickBasic code to C++.
- Jul 2006 – **Intel India Technology, Bangalore, India**
 Nov 2007 ■ **Application Developer:** Product engineering and deployment of antivirus and antispymware across the enterprise.

■ PI/Co-PI Projects/Grants Received

- **NVidia Hardware grant** for Digital Agriculture (2022).
- Recipient of **Adobe Grant of \$15000** for Systems ML research (2021).
- Co-PI for Board of Research in Nuclear Sciences (BRNS) funded project “Fire Safety Audit” (2018).

■ Research Visits (Internships)

- Aug 2017 – , *Norwegian University of Science and Technology*, Trondheim, Norway
 Oct 2017 ■ Department of Information Security and Communication Technology Faculty of Information Technology and Electrical Engineering, Trondheim, Norway
 ■ Host: Prof. Dr. Yuming Jiang
- Jan 2016 – , *Nokia Bell Labs* , Stuttgart, Germany and Dublin, Ireland
 Jun 2016 ■ Autonomous Software Systems Research, Stuttgart, Germany
 ■ Host: Prof. Dr. Volker Hilt

■ Refereed Journal Publications

- Varad Kulkarni, Nikhil Reddy, Tuhin Khare, **Abhinandan S. Prasad**, Chitra Babu, and Yogesh Simmhan, Characterizing FaaS Workflows on Public Clouds: The Good, the Bad and the Ugly In: *IEEE Transactions on Parallel and Distributed Systems*. 37(6): 1312-1329, 2026 doi:10.1109/TPDS.2026.3678606
- Sweta Dey, Rishi Raina, Sudeepta Mishra, **Abhinandan S. Prasad**, Ramesh Dharavath, GreenEdge AI: Sustainable federated learning for smart city air quality prediction In: *Elsevier Journal of Industrial Information Integration*, 50:101081, 2026. doi:10.1016/j.jii.2026.101081
- Prasad, V. G., **Abhinandan S. Prasad**, and Rao, S. A Combinatorial Auction mechanism for multiple resource procurement in cloud computing In: *IEEE Transactions on Cloud Computing*. 6(4): 904-914, 2018. doi:10.1109/TCC.2016.2541150
- **Abhinandan S. Prasad**, Koll, D., Iglesias, J. O., Aroca, J. A., Hilt, V. and Fu, X. RConf(PD): Automated resource configuration of complex services in the cloud In: *Future Generation Computer Systems* 87: 639-650, 2018. (Invited). doi:10.1016/j.future.2018.02.027
- **Abhinandan S. Prasad**, and Rao, S. A Mechanism Design Approach to Resource Procurement in Cloud Computing In: *IEEE Transactions on Computers*, 63(1): 17-30 (2014). doi:10.1109/TC.2013.106

Book Chapters

- Narayan, A., Pillai, P. S., **Abhinandan S. Prasad**, and Rao, S. Resource Procurement, Allocation, Metering, and Pricing in Cloud Computing In: *Research Advances in Cloud Computing Edited by: Sanjay Chaudhary, Gaurav Somani, Rajkumar Buyya.* 141-186 Springer isbn:978-981-10-5025-1.
doi:10.1007/978-981-10-5026-8_7

Refereed Conference Publications

- Arpit Tripathi, Abhishek Thakur, **Abhinandan S. Prasad**, Bheemarjuna Reddy Tamma, Subhrendu Chattopadhyay, MADE: An MEC Supported Platform Independent and Version Agnostic Framework for Mobile Application Deployment (Accepted) In: *IEEE ANTS 2024*
- Vrushank Ahire, Pranav Menon, Aniruddh Muley, **Abhinandan S. Prasad**, Random Adaptive Cache Placement Policy (Accepted) In: *HiPC SRS 2024*
- Varad Kulkarni, Nikhil Reddy, Tuhin Khare, Harini Mohan, Jahnvi Murali, Mohith A, Ragul B, Sanjai Balajee, Sanjjit S, Swathika, Vaishnavi S, Yashasvee, Chitra Babu, **Abhinandan S. Prasad** and Yogesh Simmhan, XFBench: A Cross-Cloud Benchmark Suite for Evaluating FaaS Workflow Platforms (Accepted with "Open Research Objects (ORO)" and "Research Objects Reviewed (ROR)" badges) (*CORE Ranking: A (CORE2020)*).
doi:10.1109/CCGrid59990.2024.00067
- Aditya Hegde, Sameer G. Kulkarni, and **Abhinandan S. Prasad**, COUNSEL: Cloud Resource Configuration Management using Deep Reinforcement Learning (Accepted with "Open Research Objects (ORO)" and "Research Objects Reviewed (ROR)" badges).
doi:10.1109/CCGrid57682.2023.00035
- Subhrendu Chattopadhyay, Sukumar Nandi, Sandip Chakraborty and **Abhinandan S. Prasad**, Amalgam: Distributed Network Control With Scalable Service Chaining In: *IFIP Networking 2020*, France.
- **Abhinandan S. Prasad**, Mayutan Arumaithurai, David Koll and Xiaoming Fu, DMC: A Differential Marketplace For Cloud Resources. In: *IEEE/ACM CCGRID 2019* (Nominated for *Best paper Award*).
doi:10.1109/CCGRID.2019.00034
- **Abhinandan S. Prasad**, David Koll, Mayutan Arumaithurai, Yuming Jiang and Xiaoming Fu, OFM: Online Fisher Market. In: *IEEE INFOCOMM 2019*.
doi:10.1109/INFOCOM.2019.8737641
- **Abhinandan S. Prasad**, Koll, D., Iglesias, J. O., Aroca, J. A., Hilt, V. and Fu, X. Optimal Resource Configuration of Complex Services in the Cloud In: *Proceedings of the 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID'17)*, Madrid, Spain, May 14-17, 2017.
doi:10.1109/CCGRID.2017.67
- **Abhinandan S. Prasad**, Arumaithurai, M., Koll, D. and Fu, X, RAERA: A Robust Auctioning Approach for Edge Resource Allocation In: *Proceedings of the Workshop on Mobile Edge Communications (MECOMM@SIGCOMM'17)*, Los Angeles, CA, USA, August 21, 2017.
doi:10.1145/3098208.3098217
- **Abhinandan S. Prasad**, Koll, D. and Fu, X. On the Security of Software-Defined Networks, In: *Fourth European Workshop on Software Defined Networks (EWSDN'15)* Bilbao, Spain, September 30 - October 2, 2015.
doi:10.1109/EWSDN.2015.70
- Prasad, V. G., Rao, S., and **Abhinandan S. Prasad** A Combinatorial Auction mechanism for multiple

resource procurement in cloud computing In: *12th International Conference on Intelligent Systems Design and Applications (ISDA'12)*, Kochi, India, November 27-29, 2012.
doi:10.1109/ISDA.2012.6416561

- o **Abhinandan S. Prasad**, and Rao, S. Citation Matching in Sanskrit Corpora Using Local Alignment In: *Sanskrit Computational Linguistics - 4th International Symposium*, New Delhi, India, December 10-12, 2010.
doi:10.1007/978-3-642-17528-2_9

■ Honours and Awards

- o IBM Champion for years 2023, 2022, and 2021 (three consecutive years).
- o The paper "Optimal Resource Configuration of Complex Services in the Cloud" invited to *Special Issue on Mobile, hybrid, and heterogeneous clouds for cyberinfrastructures* (FGCS) Journal.
- o Mar 2015: Marie Skłodowska-Curie ITN Fellowship
- o Jun 2010: Director's Merit List (IIIT Bangalore)
- o Jan 2010: Director's Merit List (IIIT Bangalore)

■ Academic Services

- o HiPC SRS 2024 Co-chair
- o *Proceedings Chair/Co-chair*: CODS-COMAD 2022, CODS-COMAD 2022, CCGRID 2023
- o *Conferences*: COMSNET 2020-2024, ANTS 2021-24
- o *Journals*:
 - IEEE Transactions on Cloud Computing
 - ACM Transactions on Autonomous and Adaptive Systems
 - Elsevier Computer Communications
 - EURASIP Journal on Wireless Communications and Networking
 - IEEE Transactions on Network and Service Management
 - IEEE Network Magazine
 - Transactions on Emerging Telecommunications Technologies
 - SADHANA